Linear systems and elimination

Exact extremizers for partial pivoting on orthogonal matrices

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Rating rationale (historical): Challenging reflects a global extremal problem with pivot-path constraints in every dimension; specialist impact concerns sharp constants on the orthogonal subclass.

Negative resolution - 2026-09-11

Solved. George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, gives an exact order-eight counterexample in the complete proof, **Theorem and Sections 1-4**. Let $\tilde{L} = L_8 + e_8 e_2^T$, changing only its $(8, 2)$ entry from -1 to 0, and take its positive-diagonal QR factor $\widetilde{Q}$. Under the stipulated first-available-row tie rule,

$$\rho_{\mathrm{PP}}(\widetilde{Q}) = \frac{5272}{63} > \sqrt{\frac{17948132}{2601}} = \rho_{\mathrm{PP}}(Q_8).$$

The proof prints exact integer-column descriptions of both orthogonal matrices and verifies all active Schur-complement maxima. This disproves the universal extremizer equality. It does not determine the true orthogonal supremum or refute the separate asymptotic leading-constant conjecture. Proof PDF · Standalone TeX.

A separate Codex-agent mathematical review returned **PASS**, with independently written rational checks of both QR conventions, all pivot ties, all 408 active entries and the positive squared growth gap. Substantial ChatGPT/Codex assistance is disclosed; this is automated-agent review, not external human peer review or formal certification. Authorship, reproducible certificates, source checks and public-branch audit. The original ID, path, statement, historical ratings and earlier checks below are retained. Peca-Medlin retains credit for the conjecture and cited element-growth analysis.

Context and notation

All elimination in this problem is in exact arithmetic. Write $\|A\|_{\max} = \max_{ij} |a_{ij}|$. A pivoting path creates successive active Schur complements $S_1 = A, S_2, \ldots, S_n$, with row/column permutations as appropriate. Its element-growth factor is

$$\rho(A) = \frac{\max_{1\le j\le n} \|S_j\|_{\max}}{\|A\|_{\max}}.$$

Partial pivoting chooses a largest-magnitude entry in the active first column; complete pivoting chooses one anywhere in the active matrix. If ties occur, a universal statement includes every admissible tie choice; a supremum includes all admissible paths. These conventions remove implementation-dependent ambiguity. Matrices are nonsingular unless otherwise stated.

Problem statement

Let L_n be the real unit lower triangular matrix whose entries strictly below the diagonal are all -1. Define Q_n by the unique QR factorization $L_n = Q_n R_n$ with positive diagonal in R_n. For Q_n, use partial pivoting with the first available row chosen in a tie. Is it true that, for every $n \ge 2$,

$$\sup_{Q\in O(n)} \rho_{\mathrm{PP}}(Q) = \rho_{\mathrm{PP}}(Q_n), \qquad O(n) = \{Q \in \mathbb{R}^{n\times n} : Q^T Q = I\}?$$

The supremum on the left includes all admissible partial-pivoting paths. The candidate is fully specified by L_n, rather than by an approximate numerical optimizer. This asks for the sharp extremizer, beyond the established exponential order of orthogonal growth.

Reference

Peca-Medlin, *Growth factors of orthogonal matrices and local behavior of Gaussian elimination with partial and complete pivoting*, published in SIAM J. Matrix Anal. Appl. (2024), §3.2 and Appendix B. The paper conjectures this equality and establishes $\rho_{\mathrm{PP}}(Q_n) = 2^{n-1}(1 + o(1))/\sqrt{3}$.

Earlier status check — 2026-09-08

Searches for *GEPP orthogonal conjecture 2026* and the exact paper title found no proof of the extremal equality. The 2026 butterfly paper still identifies orthogonal partial-pivoting growth as open. The August Shah–Urschel results concern different growth questions and pivot strategies; their exponential examples do not establish this exact supremum.

Audit update — 2026-09-10

Rechecked Peca-Medlin's manuscript, §3.2 and Appendix B, and its SIAM publication, SIAM J. Matrix Anal. Appl. 45 (2024), 1599–1620. The candidate extremizers remain supported by the stated construction and experiments, without a general optimality proof. Searches for subsequent orthogonal GEPP extremizer results found no resolution.